



📅 23 Dec 2025

Slide 1

Using qualitative causal mapping
to visualise
how people understand complex social systems
from large quantities of interview or report data

STIG (Systems Thinking Interest Group)
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Outline

Context and our motivation

What is causal mapping?

Extended case study – central bankers' perceptions of the economy

Where to go with this? Q&A

Key messages

1. A useful step to understanding complex social systems is to collect and analyse multiple stakeholders' perceptions about what is driving change
2. Causal mapping of large digitised natural language data can help us to do this better
3. It does this by enabling us to identify, combine and visualise stakeholders' mental models of the system more systematically

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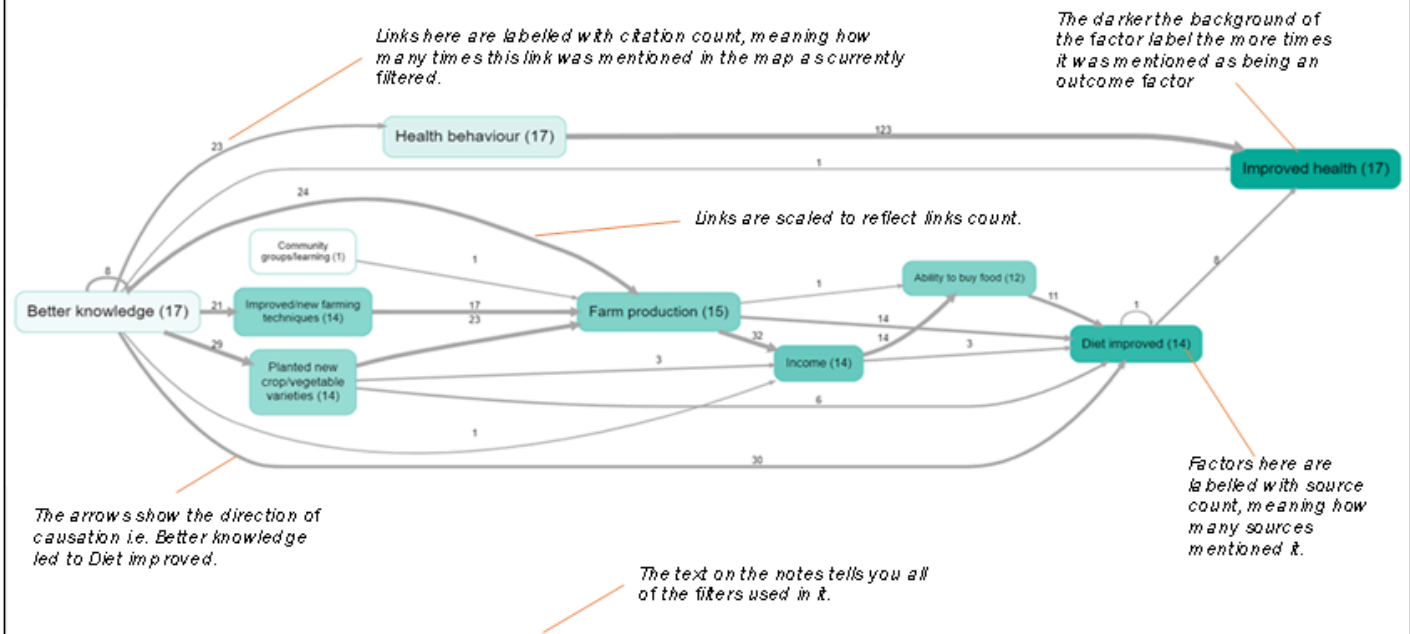
Automated coding of an individual speech

Individual text, highlighting passages identified and coded by the AI (143 for this source)

Individual causal map (part)

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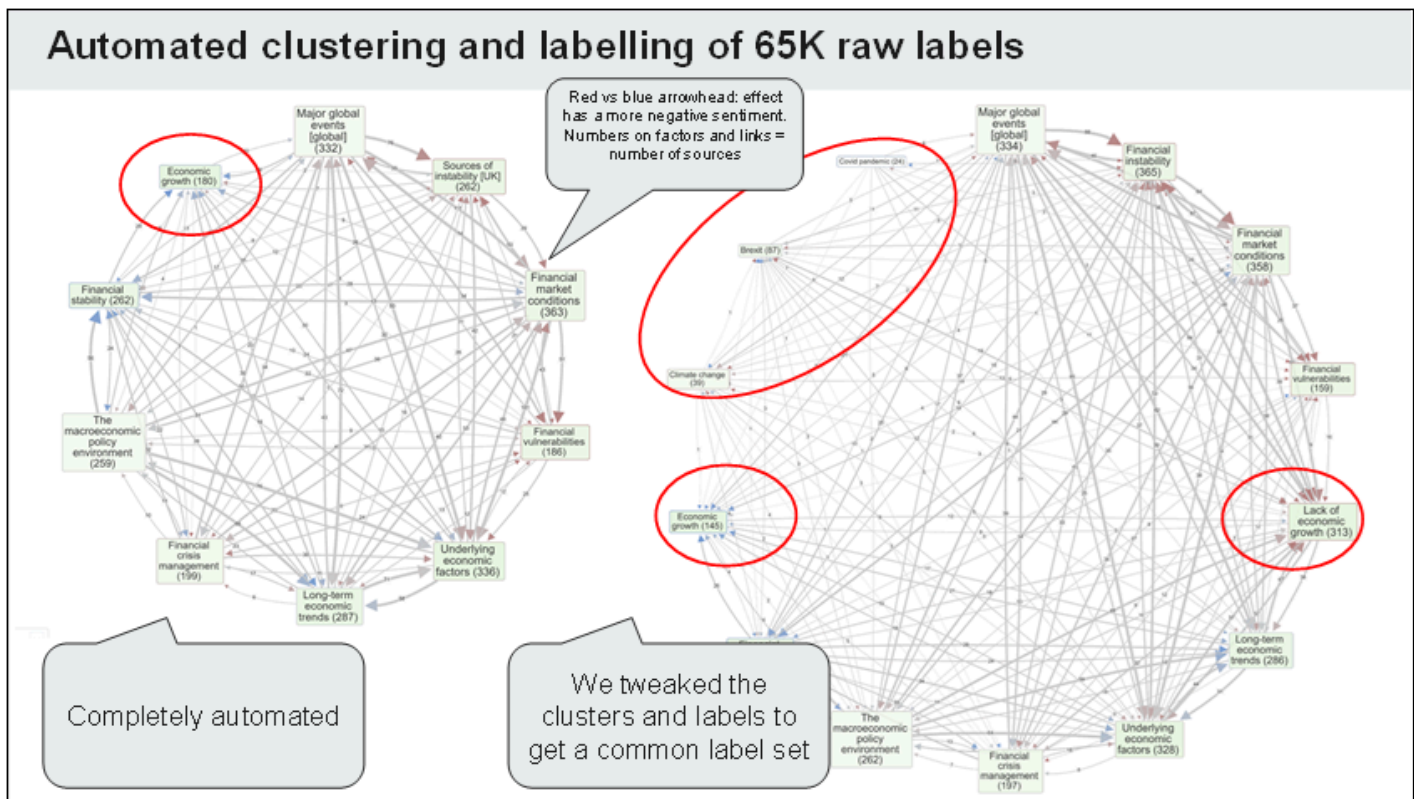
How to read a causal map



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**Result of raw coding:
A causal knowledge
graph with 60,000
causal factor labels**

****Causal mapping identifies the factors people use in explanations: often difficult to reconcile with statistical variables**

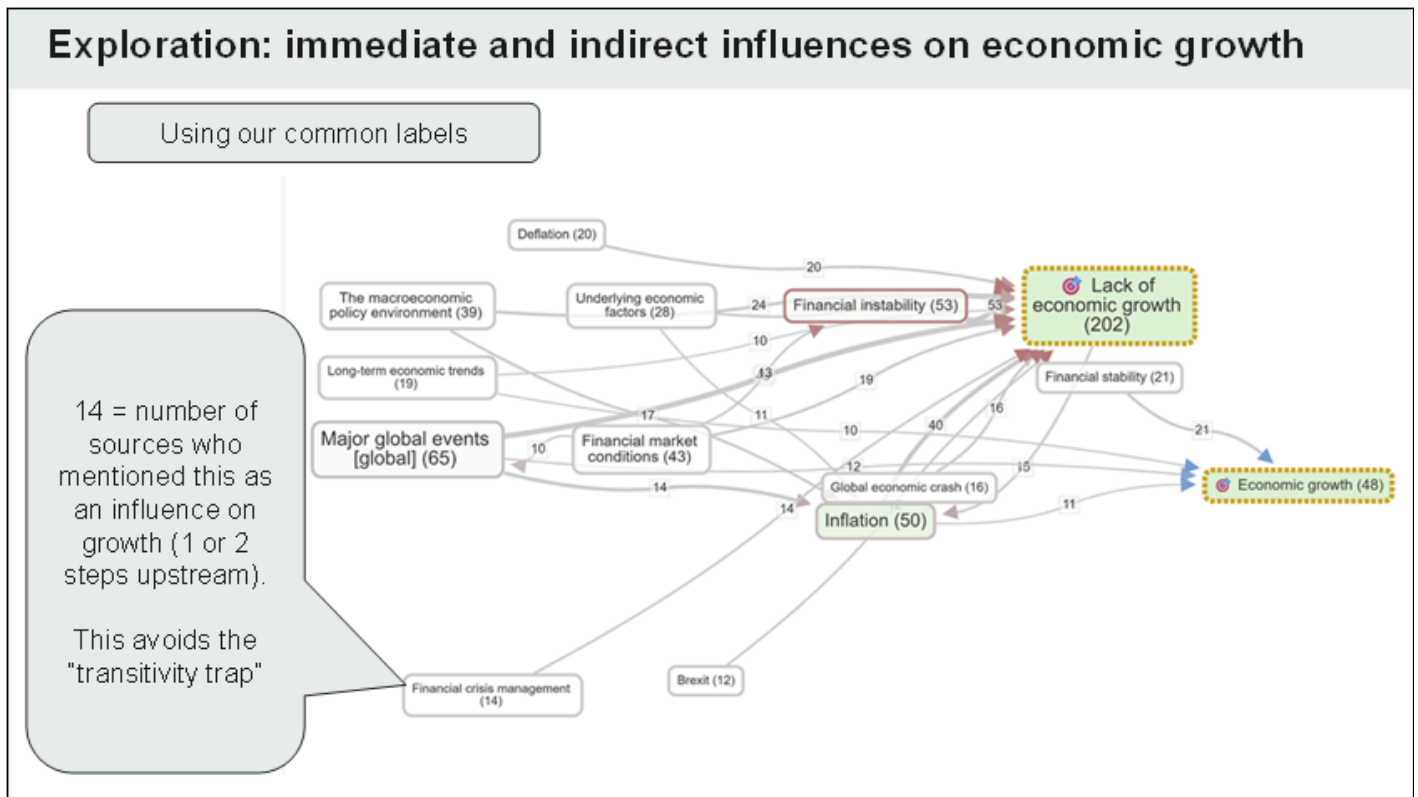


What is the opposite of inflation?

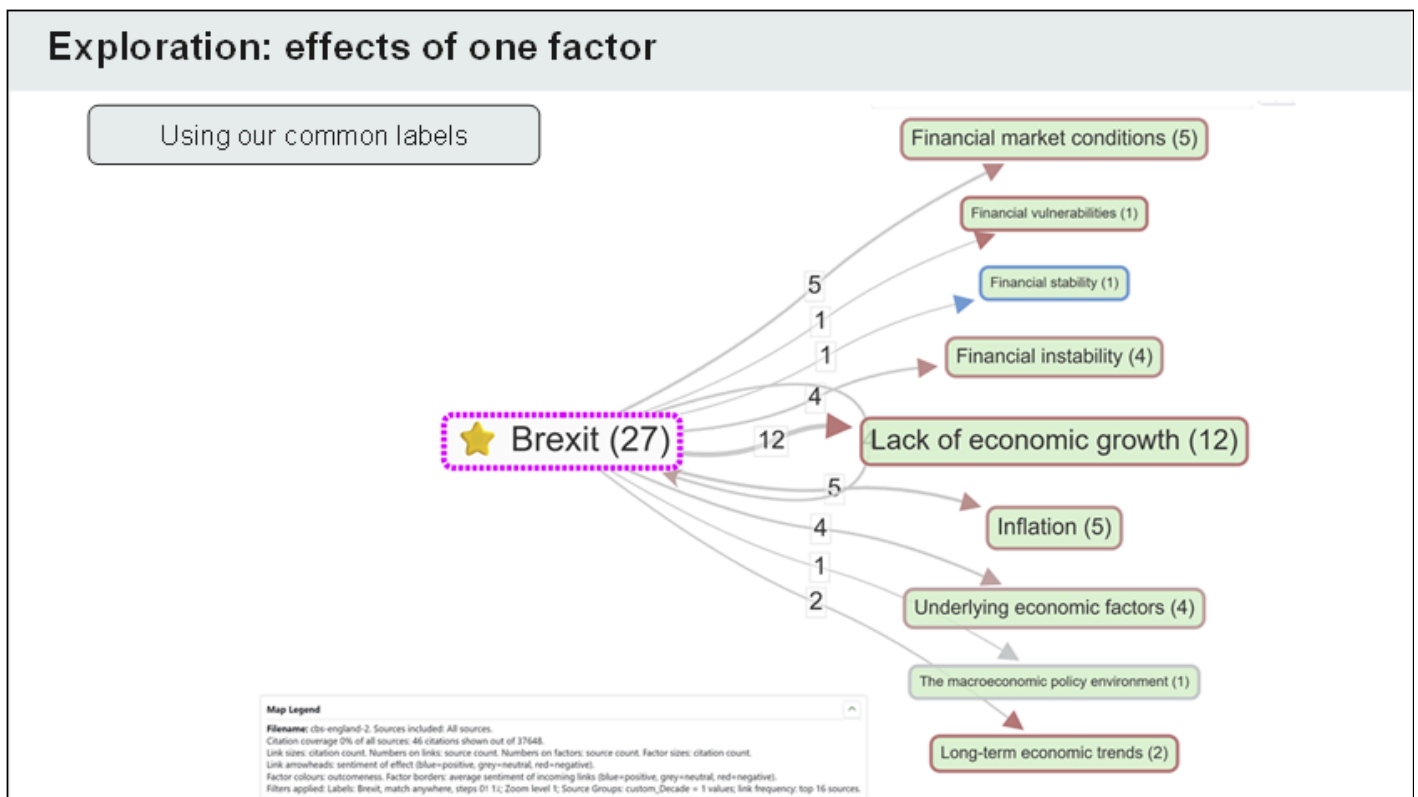
Cognitively, “Inflation” and “Deflation” and “Low inflation” are each attributed with *overlapping but different* causal powers. These powers are not well captured as a single “variable”

**Causal mapping identifies the factors people use in explanations: often difficult to reconcile with statistical variables

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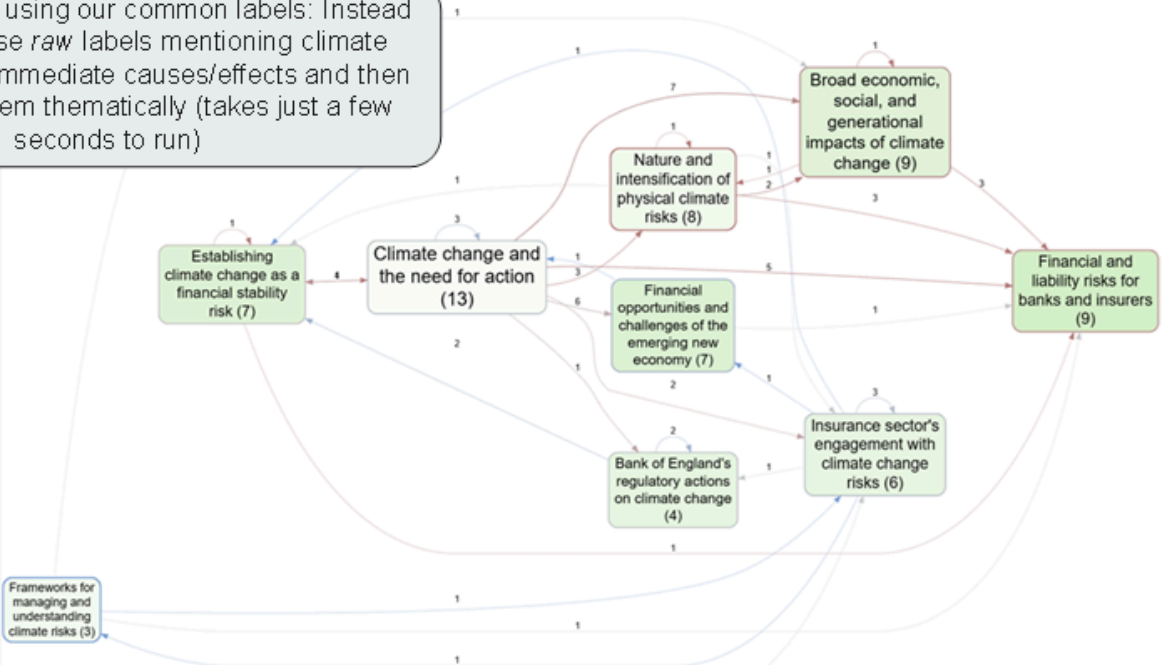
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What are the narratives around climate change

Here we are not using our common labels: Instead we isolate those *raw* labels mentioning climate change and its immediate causes/effects and then cluster/label them thematically (takes just a few seconds to run)



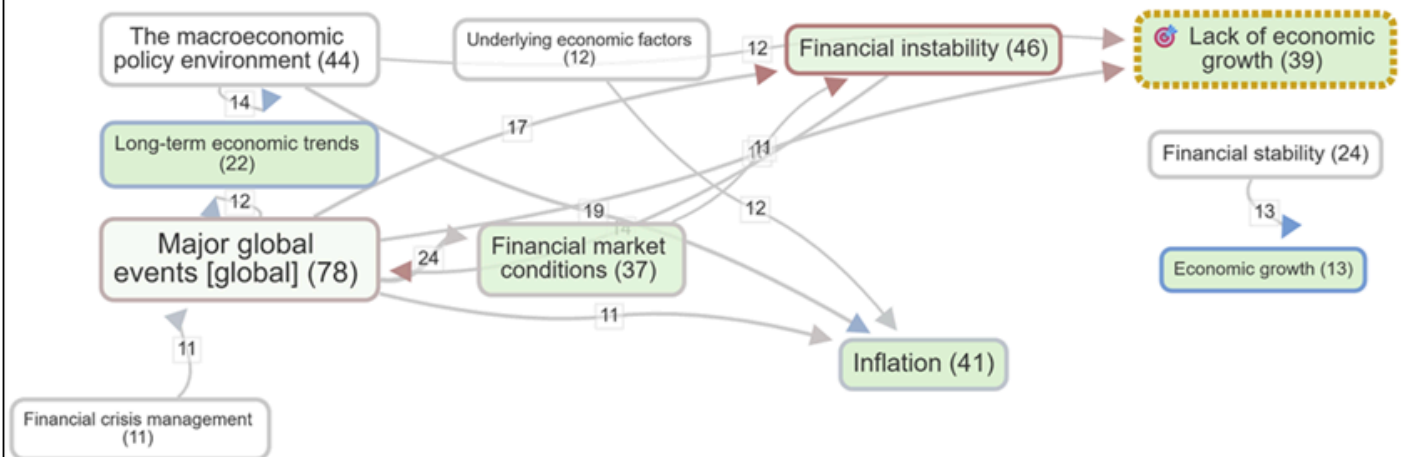
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Differences by decade in explanations of lack of growth

Keeping the common labels across all 3 decades

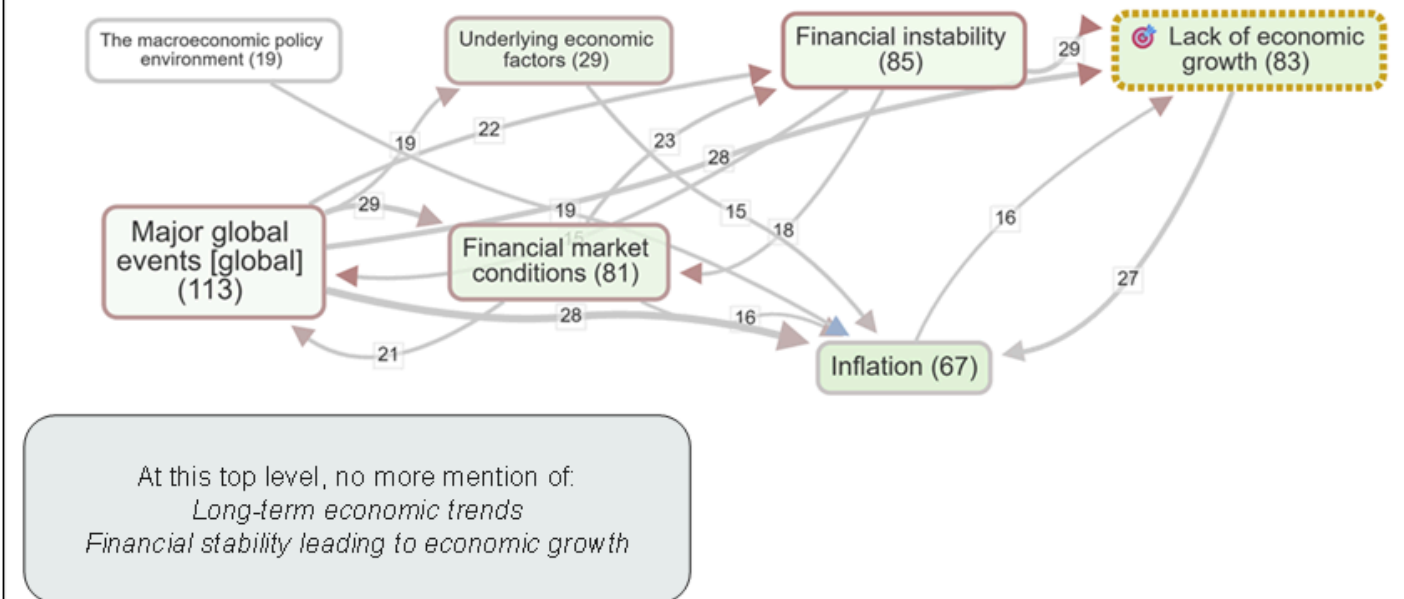
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1996-2005



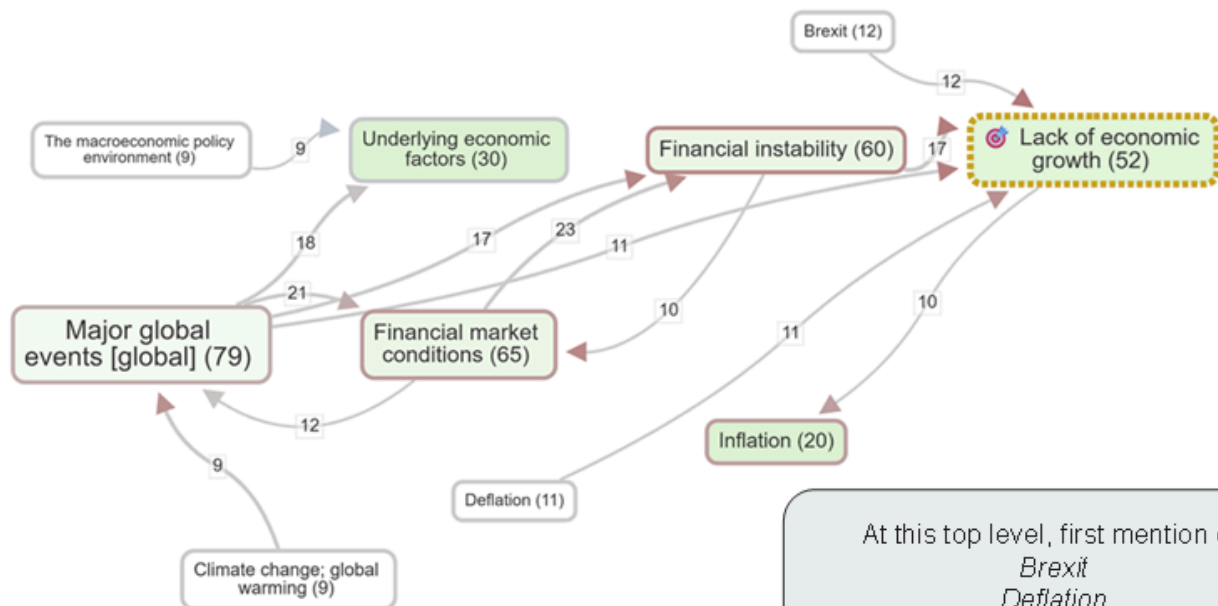
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2006-2015



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2016-2023



At this top level, first mention of:
Brexit
Deflation
Climate change

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That 3-decade comparison would have shown much bigger differences if we had used factors specific to each decade rather than the 3 decades together

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Where to go with this? Q&A

Thank you!

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Annex (not shown)

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Raw labels and cluster labels in “meaning space”

Raw cause/effect
labels from coding

Cluster labels

Clustering raw labels
and finding cluster
labels: half art, half
science

